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Macroeconomic Conditions and Capital Raising

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1 Big picture

- **Question.** Do macroeconomic conditions affect firms' ability to raise capital, the type of capital they raise, and the contractual structure of the securities they issue?
- **Scope.** The paper studies a broad set of U.S. corporate financing events from 1971 to 2007: public debt, seasoned equity offerings, convertible bonds, syndicated bank loans, and private placements of equity and debt.
- **Core distinction.** The paper separates *demand-side* explanations from *supply-side* explanations. Demand-side theories emphasize changes in firms' desired financing and asymmetric-information costs. Supply-side theories emphasize credit crunches, intermediary capital, and investor flight to quality.
- **Main result.** Macroeconomic conditions matter for both the quantity and composition of corporate capital raising.
- **Credit-quality heterogeneity.** Noninvestment-grade borrowers face procyclical access to capital markets: they raise less capital in downturns. Investment-grade borrowers raise public debt countercyclically and are much less likely to be shut out.
- **Security-choice result.** For weaker firms, public equity and low-rated public debt are especially cyclical. For stronger firms, public bond issuance increases in downturns.
- **Contract-design result.** In poor macroeconomic conditions, issued debt shifts toward safer structures: shorter maturity and, in some specifications, more security/collateral.
- **Use-of-proceeds result.** Investment-grade firms appear to raise capital during downturns partly to preserve liquidity and save proceeds as cash. Lower-quality firms are more likely to use capital immediately and are less able to build cash buffers.
- **Broad interpretation.** The data support a supply-of-capital channel. Downturns and weak credit markets shift available financing toward high-credit-quality firms and safer securities.
- **Economic takeaway.** Financial flexibility and credit quality matter because macroeconomic downturns change not only the price of external capital but also whether firms can access capital and what kind of contracts investors are willing to buy.

2 Data and measurement

2.1 Capital-raising sample

- The sample includes U.S. industrial firms that have corresponding accounting information in Compustat as of the fiscal year-end immediately prior to issuance.
- The sample contains six major external financing categories:
 - seasoned equity offerings (SEOs);
 - convertible bonds;
 - other public debt;
 - private syndicated loans;
 - private placements of equity;
 - private placements of debt.
- Public securities are observed over 1971–2007.

- Private loans are observed over 1988–2007.
- Private placements are observed over 1981–2007.
- The primary sources not covered are regular bilateral bank loans and commercial paper.

Interpretation: The sample is unusually broad. It allows the paper to study not only whether firms raise capital, but how they substitute across public securities, private loans, and private placements.

2.2 Security-type data sources

- SEOs, convertibles, public bonds, and private placements come from securities issuance databases such as SDC and Mergent FISD.
- Syndicated loans come from DealScan for the later part of the sample.
- Firm characteristics come from Compustat.
- Public bond ratings and contract features such as maturity, secured status, and credit quality come from Mergent FISD.

Interpretation: Combining multiple issuance datasets is necessary because the paper’s key question concerns substitution across financing sources.

2.3 Macroeconomic condition measures

The paper uses three main measures of poor macroeconomic or credit conditions:

1. **NBER recession dummy:** equals one in months designated as recession months by the NBER.
2. **Low GDP growth dummy:** equals one in quarters in which GDP growth is below the 25th percentile of quarterly GDP growth over 1971–2007.
3. **Weak credit dummy:** equals one when the net percentage of senior loan officers tightening standards for loans to large and medium-sized firms is positive.

Interpretation: The three measures capture related but distinct states: broad recession, low real activity growth, and direct credit supply tightness.

2.4 Firm-level controls

The main firm controls are:

- firm age;
- log total assets;
- market leverage;
- market-to-book ratio;
- fixed-asset ratio;
- cash flow;
- cash;
- inverse interest coverage;
- debt-rating dummy;

- sales growth;
- prior twelve-month stock return.

Interpretation: These variables capture demand for capital, collateral capacity, firm size, investment opportunities, financial constraints, and credit quality.

2.5 Issue and contract variables

- **Short-term bond/loan dummy:** equals one if proceeds-weighted initial maturity is less than five years.
- **Secured bond/loan dummy:** equals one if proceeds-weighted secured status exceeds a threshold.
- **Credit quality categories:** public bonds are grouped into ratings buckets, including A's, investable B's, speculative B's, C's, and nonrated debt.
- **Cash accumulation measure:** the paper studies how much cash and short-term investments rise after capital raising.

Interpretation: These variables let the paper ask whether downturns alter the *structure* of external finance, not just its amount.

3 Models and empirical specifications

3.1 Aggregate and normalized issuance comparisons

The descriptive analysis computes, for each month, the share of capital raised through each financing method:

$$Share_{m,t} = \frac{Proceeds_{m,t}}{\sum_k Proceeds_{k,t}}, \quad (1)$$

where m indexes financing methods such as SEOs, public bonds, loans, and private placements.

Interpretation: Comparing shares rather than raw dollars makes it possible to study substitution across financing channels within a month.

3.2 Multinomial logit model of security choice

The main model of security choice is a multinomial logit:

$$Pr(security\ type = j) = \frac{e^{\beta_j' X}}{\sum_{k=0}^6 e^{\beta_k' X}}, \quad (2)$$

where:

- $j = 0$ means no security issue;
- $j = 1$ is a bank loan;
- $j = 2$ is a public bond;
- $j = 3$ is a convertible bond;
- $j = 4$ is an SEO;
- $j = 5$ is a private placement of equity;
- $j = 6$ is a private placement of debt;
- β_0 is normalized to zero.

Interpretation: Each coefficient compares the probability of issuing a given security to not issuing any security, controlling for firm characteristics and macroeconomic conditions.

3.3 Specification variants

The paper estimates variants using different macroeconomic-condition variables:

$$Pr_j = f(X_{firm}, TermSpread, RecessionDummy, IndustryFE), \quad (3)$$

$$Pr_j = f(X_{firm}, TermSpread, LowGrowthDummy, IndustryFE), \quad (4)$$

$$Pr_j = f(X_{firm}, TermSpread, WeakCreditDummy, IndustryFE). \quad (5)$$

Interpretation: Robustness across these three measures indicates that results are not driven only by the formal NBER recession classification.

3.4 Credit-quality subsamples

The multinomial logit model is estimated separately for:

- all firms;
- noninvestment-grade firms;
- investment-grade firms.

The paper classifies noninvestment-grade firms as those that never issued an investment-grade public bond during the sample period based on Moody's ratings.

Interpretation: The split is central. Supply-side theories predict that downturns affect low-quality borrowers differently from high-quality borrowers.

3.5 Cash-holdings/use-of-proceeds regression

The paper estimates regressions of the form:

$$Y = \beta_0 + \beta_1 \ln \left[\left(\frac{capital\ raised}{total\ assets} \right) + 1 \right] + \beta_2 \ln \left[\left(\frac{capital\ raised}{total\ assets} \right) + 1 \right] \times RecessionDummy + \beta_3 \ln(total\ assets) + \epsilon, \quad (6)$$

where Y is the change in cash and short-term investments relative to assets over horizons of 1, 4, or 8 quarters.

Interpretation: The interaction coefficient β_2 asks whether proceeds raised during recessions are saved as cash rather than immediately spent.

3.6 Debt maturity and security logit models

For public bonds and bank loans, the paper estimates logit models for contract design:

$$Pr(ShortTerm_i = 1) = \Lambda(X_i' \gamma), \quad (7)$$

$$Pr(Secured_i = 1) = \Lambda(X_i' \delta), \quad (8)$$

where X_i includes firm controls, the term spread, and the macroeconomic-condition dummy.

Interpretation: These models test whether downturns and credit tightness make debt safer from investors' perspective by shortening maturity or increasing collateralization.

4 Identification and empirical design

4.1 What identifies the empirical patterns

- The paper combines time-series variation in macroeconomic conditions with firm-level variation in credit quality and financing needs.
- It compares issue probabilities and security structures across expansions, recessions, high- and low-GDP-growth quarters, and weak-credit quarters.
- It controls for firm characteristics that may proxy for capital demand, collateral, investment opportunities, and financial health.

Interpretation: The identification is not a randomized experiment. It is a large-sample conditional comparison across macro states, firm types, and financing instruments.

4.2 Demand-side versus supply-side mechanisms

- Demand-side asymmetric-information models predict that poor conditions should lead firms to issue less information-sensitive securities, shifting away from equity toward safer securities.
- Supply-side credit-crunch and flight-to-quality models predict that downturns reduce capital supply to weak borrowers and shift investor demand toward high-quality and safer securities.
- The paper distinguishes these channels by comparing investment-grade and noninvestment-grade firms, high- and low-rated bonds, proceeds saved as cash, and debt contract design.

Interpretation: A pure demand story cannot easily explain why high-quality borrowers issue public debt countercyclically while lower-quality borrowers lose access.

4.3 Why credit quality is central

- Noninvestment-grade firms are more likely to face financing constraints during downturns.
- Investment-grade firms are more likely to be viewed as safe assets during flight-to-quality episodes.
- If supply shifts toward safety, issuance by high-quality borrowers should increase or remain available during downturns, while issuance by low-quality borrowers should decline.

Interpretation: The credit-quality split is the paper's main lever for interpreting the results as supply-side rather than only demand-side.

4.4 What the paper cannot fully prove

- The paper cannot directly observe latent demand for capital by firms that do not issue.
- It cannot perfectly separate borrower demand from investor supply in every issuance market.
- Private bilateral bank loans and commercial paper are not fully observed.
- The credit-quality classification is based on observed bond ratings and issuance histories.
- The evidence is strongest as a set of mutually reinforcing patterns rather than as one clean causal experiment.

Interpretation: The paper's strength is breadth: multiple instruments, macro measures, credit-quality splits, uses of proceeds, and contract features point in the same direction.

5 Tables and figures

This section uses the same *Read* plus *Economic message* structure. The visuals are directly cropped from the paper and grouped compactly.

5.1 Table 1 and Figure 1: overview of capital raised over time

Read:

- Table 1 reports average monthly proceeds in constant 2000 dollars by capital-raising method across NBER expansion and recession subperiods.
- SEOs tend to be lower during recessions, while public bond proceeds rise substantially in several downturns.
- Figure 1 shows moving-average time-series trends in proceeds for SEOs, convertibles, public debt, private loans, and private placements.
- Public debt rises markedly during some recessions, while equity issuance tends to fall.

Economic message: Aggregate capital raising is not uniformly procyclical. The composition of financing changes across the business cycle.

Table 1
Sample descriptive statistics

	Average Proceeds per Month (constant 2000 US\$ million)					
	SEOs	Convertibles	Public Bonds	Private Loans	Private Placements Equity	Debt
January 1971–October 1973	395.3	29.5	936.1	–	–	–
November 1973–March 1975 (recession)	137.2	16.9	1,632.1	–	–	–
April 1975–December 1979	308.0	35.8	1,096.3	–	–	–
January 1980–July 1980 (recession)	395.1	42.0	2,337.7	–	–	–
August 1980–June 1981	1,085.1	141.3	1,626.2	–	–	–
July 1981–November 1982 (recession)	413.6	76.7	1,498.2	–	55.3	477.3
December 1982–June 1990	710.9	477.6	3,905.5	11,915.9	22.6	526.4
July 1990–March 1991 (recession)	391.9	955.3	4,000.6	7,504.1	180.9	2,167.5
April 1991–February 2001	2,080.5	1,984.5	16,451.4	26,078.9	221.7	815.0
March 2001–November 2001 (recession)	1,677.9	8,264.4	35,598.8	48,832.1	503.2	766.8
December 2001–December 2007	1,680.4	4,948.9	19,623.0	38,876.3	532.3	569.1
All	844.0	1,545.0	8,124.2	12,291.6	155.5	718.8

The sample includes all SEOs, convertible bonds, other public debt, private loans, and private placements of equity and debt issued by U.S. industrial firms that have corresponding accounting information in Compustat as of the fiscal year-end immediately prior to the issue. Sample period is between 1971 and 2007, except for private loans (1988–2007) and private placements (1981–2007). We divide the sample into six expansion periods and five recession periods based on the NBER classification. For each subperiod, we report the averages of proceeds raised per month in constant 2000 US\$ for each of the six security types.

Table 1: sample descriptive statistics

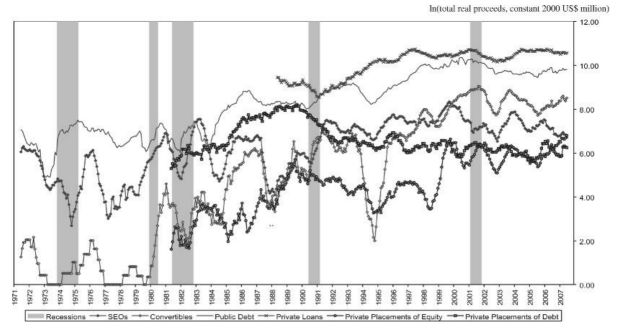


Figure 1
Proceeds raised from different types of securities over time

This figure presents the log of proceeds raised in real terms (constant 2000 US\$ million) by types of security issues for each calendar month from 1971 to 2007. To smooth out the series, we plot the eleven-month moving averages around a specific calendar month. The shaded areas correspond to recessions as defined by the NBER.

Figure 1: proceeds by security type over time

5.2 Table 2: macroeconomic conditions and security-issue shares

Read:

- Panel A covers 1971–1987, before private loans are available in DealScan.
- In Panel A, SEOs are lower in recessions than in expansions, while public bonds are higher.
- Panel B covers 1988–2007, when loans are included.
- In Panel B, public loans dominate capital raised, but loan shares are lower during recessions and weak-credit periods.
- Public bonds and convertibles tend to account for a larger fraction of capital raised in downturns.

Economic message: Firms substitute across financing sources as macroeconomic and credit conditions change. The substitution pattern is consistent with supply shifting toward safer public debt and away from lower-quality financing.

Table 2
Macroeconomic conditions and security issues

Panel A: 1971–1987

	Number of months	Average of Relative Proceeds within Month (%)					
		SEOs	Convertibles	Public Bonds	Private Placements		
					Equity	Debt	
Expansion	162	23.4	3.4	63.1	1.9	22.7	3.9
Recession	41	15.2	2.1	70.5	1.9	27.4	7.8
t (difference)		-2.22	-1.32	1.66	0.03	0.92	2.25
High GDP growth	144	22.7	3.6	61.5	1.8	23.5	4.8
Low GDP growth	59	19.3	2.1	71.9	2.2	24.0	0.81
t (difference)		-1.08	-1.58	2.64	0.27	0.08	

(continued)

Table 2, Panel A: 1971–1987

Table 2
Continued

Panel B: 1988–2007

	Number of months	Average of Relative Proceeds within Month (%)					
		SEOs	Convertibles	Public Bonds	Public Loans	Private Placements	
						Equity	Debt
Expansion	222	3.8	5.1	28.5	58.0	0.7	3.9
Recession	18	2.2	7.6	31.9	49.4	1.0	7.8
t (difference)		-1.91	2.27	1.45	-2.22	0.3	2.25
High GDP growth	180	3.9	5.0	28.6	57.8	0.7	4.0
Low GDP growth	60	3.1	5.9	29.4	55.9	0.9	4.8
t (difference)		-1.71	1.25	0.39	-0.79	0.2	0.81
Strong Credit Supply	108	3.8	4.0	26.0	64.1	0.5	1.6
Weak Credit Supply	105	4.1	7.0	33.4	51.3	0.8	3.3
t (difference)		0.69	5.09	4.16	-6.54	0.3	3.18

This table presents the averages of relative proportions of proceeds raised through six types of securities within each calendar month. The sample includes all SEOs, convertible bonds, other public debt, private loans, and private placements of both equity and debt issued by U.S. industrial firms that have corresponding accounting information in Compustat as of the fiscal year end immediately prior to the issue. Sample period is between 1971 and 2007 for SEOs, convertibles, and other public bonds; 1988–2007 for private loans; and 1981–2007 for private placements. Expansions and recessions are based on the NBER classification. A month is defined as low growth if GDP growth in that particular quarter is below the 25th percentile of economic growth over the entire sample period. A month with weak credit supply takes a value of one if the net percentage of senior loan officers tightening standards for loans to large and medium firms is positive for that particular quarter. This classification is based on a Federal Reserve survey available since the 2nd quarter of 1990. For each calendar month, we first calculate the relative proportions of each of the six security types within that month. Panel A reports the monthly averages between 1971 and 1987 where private placements are treated as missing prior to 1981, while Panel B reports the corresponding numbers from 1988 to 2007, respectively.

Table 2, Panel B: 1988–2007

5.3 Table 3 and Figure 2: public debt by credit quality

Read:

- Table 3 decomposes public debt proceeds by maturity, security, rating status, and credit quality.
- During recessions, short-term public debt becomes a larger share of public debt issuance.
- The share of C-rated and nonrated issues falls in downturns, while the share of A-rated debt rises.
- Figure 2 shows that proceeds raised through low-rated public bonds fall sharply in recessions, whereas high-quality public bonds remain relatively strong.

Economic message: Public debt issuance is not countercyclical for all firms. It is countercyclical for safer firms and much more fragile for low-credit-quality issuers.

	Number of months	Average of Relative Proceeds within Month (%)							
		Short-term	Secured	Nonrated	C's (C to Caa)	Speculative B's	Investable B's	A's (A3 to Aaa)	
Expansion	381	13.2	5.3	6.2	3.7	25.8	18.7	45.6	
Recession	58	22.9	4.8	2.7	1.2	14.4	15.7	66.0	
t (difference)		2.80	-0.24	-2.12	-2.57	-3.97	-1.35	5.70	
High GDP growth	321	15.1	4.8	6.5	3.8	26.2	18.6	45.0	
Low GDP growth	118	16.3	6.8	3.8	2.4	19.0	17.7	57.2	
t (difference)		1.53	1.65	-2.09	-1.88	-3.31	-0.52	4.40	

	Number of months	Average of Relative Proceeds within Month (%)							
		Short-term	Secured	Nonrated	C's (C to Caa)	Speculative B's	Investable B's	A's (A3 to Aaa)	
Expansion	195	14.3	5.9	3.1	2.8	32.8	23.2	38.0	
Recession	18	22.9	0.7	1.2	0.5	14.5	26.8	56.9	
t (difference)		2.45	-3.01	-1.87	-2.60	-4.76	1.12	3.96	
High GDP growth	156	14.2	4.0	3.2	2.8	34.2	22.6	37.2	
Low GDP growth	37	17.1	2.7	2.3	2.1	23.2	26.1	46.3	
t (difference)		1.28	-1.85	-1.51	-1.35	-4.51	1.74	3.02	
Strong Credit Supply	108	11.2	3.8	3.6	3.4	36.4	22.8	33.7	
Weak Credit Supply	105	18.9	3.5	2.3	1.8	25.9	24.3	45.7	
t (difference)		3.98	-0.61	-2.35	-3.41	-4.94	0.87	4.54	

This table presents the averages of relative proportions of proceeds raised through public debt issues with various characteristics. Short-term months are those firm-months with proceeds-weighted initial maturities shorter than or equal to five years. Short-term debt is only available since 1985. Secured months are those firm-months with proceeds-weighted issue-level secured dummy greater than or equal to 0.5. We group all public debt into five categories based on credit ratings from Moody's: not rated, C's (C to Caa), speculative B's (B3 to Ba1), investable B's (Baa3 to Baa1), and A's (A3 to Aaa). Expansions and recessions are based on the NBER classification. A month is defined as low growth if GDP growth in that particular quarter is below the 25th percentile of economic growth over the entire sample period. A month is defined as weak credit supply if the net percentage of senior loan officers tightening standards for loans to large and medium firms is positive for that particular quarter, and is based on Federal Reserve survey. For each calendar month, we first calculate the relative proportions of each of the bond types out of total proceeds raised from public debt within that month. Panel A reports the monthly averages for the full sample period, while Panel B reports the monthly averages since the 2nd quarter of 1990, when the Federal Reserve survey became available.

Table 3: types of public debt issues

5.4 Table 4: firm characteristics by security type

Read:

- Issuing firms are larger, older, and have stronger prior stock performance than nonissuing firms.
- Equity issuers tend to be smaller, younger, and have higher market-to-book ratios.
- Public debt issuers are larger and have higher fixed-asset ratios than other issuers.
- Private loan and private debt placement issuers tend to be smaller than public debt issuers.

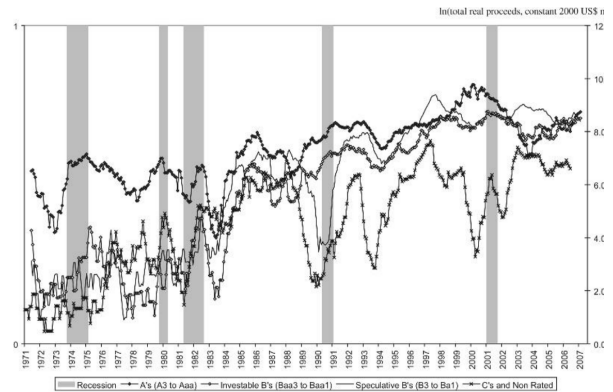


Figure 2
Proceeds raised from public bonds by credit quality over time

This figure presents the log of proceeds raised in real terms (constant 2000 US\$ million) by public bonds of various quality for each calendar month from 1971 to 2007. To smooth out the series, we plot the eleven moving average around a specific calendar month. The shaded areas correspond to recessions as defined

Figure 2: public bonds by credit quality

Economic message: Different financing instruments are used by different types of firms. The multivariate analysis must therefore control for firm characteristics before interpreting macroeconomic effects.

Table 4
Firm characteristics by security issues: Univariate analysis

	Averages per Firm-Months Observations							
	No Issue	Issue	SEOs	CBs	Bonds	Loans	Private Placements	
							Equity	Debt
Firm Age	13.283	16.229	9.862	15.963	21.997	16.289	9.804	17.932
Log(Total Assets)	4.541	5.700	4.337	6.076	7.520	5.492	3.214	6.390
Leverage	0.278	0.323	0.278	0.305	0.395	0.321	0.247	0.331
Market to Book	1.711	1.751	2.468	2.199	1.261	1.443	3.860	1.356
Fixed Asset Ratio	0.308	0.344	0.316	0.275	0.429	0.322	0.242	0.419
Cash Flow	-0.028	-0.002	-0.010	0.014	0.070	0.036	-0.507	0.017
Cash	0.155	0.120	0.187	0.209	0.059	0.093	0.324	0.078
Inverse Interest Coverage	0.182	0.235	0.183	0.155	0.306	0.245	0.039	0.300
Rated Firm Dummy	0.164	0.439	0.163	0.547	0.834	0.409	0.087	0.394
Sales Growth	0.180	0.268	0.500	0.362	0.191	0.232	0.299	0.186
Stock Return	0.174	0.357	0.970	0.582	0.221	0.193	0.416	0.174
Term Spread	1.065	1.214	1.158	1.325	1.242	1.186	1.402	1.221
Recession Dummy	0.109	0.078	0.074	0.078	0.087	0.068	0.081	0.108
Low Growth Dummy	0.248	0.215	0.194	0.222	0.215	0.213	0.258	0.223
Weak Credit Dummy	0.490	0.472	0.437	0.526	0.482	0.464	0.518	0.489
N	1,585,431	45,980	7,170	2,546	10,400	20,322	2,957	4,547

This table presents the averages of firm characteristics for each non-issuing month as well as issuing months over the entire sample period. We also report the results separately for six security types. These characteristics are firm age, natural logarithm of the total assets, market leverage, market-to-book, fixed asset ratio, cash flow, cash, the inverse of interest coverage, a debt-rating dummy, sales growth, stock return, and the term spread. Inverse interest coverage is defined as the natural logarithm of $(1 + \text{interest/EBIT})$, and stock return is calculated over the previous twelve months. Detailed definition of each characteristic is provided in the Appendix Table A1. Sample period is between 1971 and 2007 for SEOs, convertibles, and other public bonds, 1988–2007 for private loans, and 1981–2007 for private placements.

5.5 Table 5: multinomial logit model of security choice

Read:

- Panel A estimates the model for all firms, with no security issue as the base outcome.
- For all firms, recession and low-GDP-growth coefficients are negative for SEOs, consistent with procyclical public equity issuance.
- Panel B shows that noninvestment-grade firms are much more likely to experience declines in public equity and lower-quality debt issuance during downturns.
- Panel C shows that investment-grade firms are different: public bond issuance is more available, and in some specifications more likely, in downturns.
- Firm controls behave as expected: larger firms are more likely to issue public debt; higher market-to-book firms are more likely to issue equity; prior stock returns are positively related to equity issuance.

Economic message: The cyclicity of capital raising depends strongly on borrower quality. Downturns restrict low-quality firms but do not similarly restrict investment-grade borrowers.

Table 5
A multinomial logit model of security choice

Panel A: All Firms	(1)								(2)							
	Private Placements				Private Placements				Private Placements				Private Placements			
	Loan	Bond	Current	SEO	Equity	Debt	Loan	Bond	Current	SEO	Equity	Debt	Loan	Bond	Current	SEO
Firm Age	-0.001***	0.007***	-0.010***	-0.004***	-0.009***	-0.003***	-0.003***	0.007***	-0.010***	-0.001***	-0.009***	-0.006***	-0.001***	-0.001***	-0.001***	-0.001***
Log(Total Assets)	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Market Leverage	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Market-to-Book	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Fixed Asset Ratio	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Cash Flow	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Cash	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Inverse Interest Coverage	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Debt Rating Dummy	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Sales Growth	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Stock Return	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Term Spread	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Recession Dummy	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Low Growth Dummy	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Constant	-4.120***	-0.722***	-8.806***	-5.118***	-7.426***	-8.911***	-4.120***	-0.902***	-8.902***	-5.107***	-7.401***	-8.911***	-4.120***	-0.902***	-8.902***	-5.107***
Observations	728,639				728,639				728,639				728,639			
Pseudo R ²	0.001				0.001				0.001				0.001			

(continued)

Table 5
Continued

Panel B: Noninvestment-Grade Firms	(1)								(2)							
	Private Placements				Private Placements				Private Placements				Private Placements			
	Loan	Bond	Current	SEO	Equity	Debt	Loan	Bond	Current	SEO	Equity	Debt	Loan	Bond	Current	SEO
Firm Age	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***	-0.001***
Log(Total Assets)	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Market Leverage	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***
Market-to-Book	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Fixed Asset Ratio	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***
Cash Flow	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Cash	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***
Inverse Interest Coverage	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Debt Rating Dummy	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Sales Growth	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Stock Return	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Term Spread	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	0.001***
Recession Dummy	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***
Low Growth Dummy	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***	0.001***	0.001***	-0.001***	0.001***	0.001***	0.001***
Constant	-4.267***	-0.427***	-8.102***	-5.954***	-7.260***	-8.126***	-4.267***	-0.501***	-8.102***	-5.954***	-7.260***	-8.126***	-4.267***	-0.501***	-8.102***	-5.954***
Observations	616,144				616,144				616,144				616,144			
Pseudo R ²	0.072				0.072				0.072				0.072			

(continued)

Table 5

Continued

Panel C: Investment Grade Firms

	(1)								(2)							
					Private Placements								Private Placements			
	Loan	Bond	Convert	SEO	Equity	Debt			Loan	Bond	Convert	SEO	Equity	Debt		
Firm Age	0.00629*** (0.00230)	0.00546* (0.00185)	-0.00799 (0.00582)	-0.0251*** (0.00571)	-0.0385*** (0.00946)	-0.0170*** (0.00457)	0.00624*** (0.00236)	0.00533* (0.00315)	-0.00907 (0.00577)	-0.0252*** (0.00570)	-0.0384*** (0.00940)	-0.0173*** (0.00454)				
ln(Total Assets)	0.168*** (0.0231)	0.460*** (0.0387)	0.185*** (0.0479)	-0.135*** (0.0501)	0.345*** (0.124)	0.311*** (0.0387)	0.168*** (0.0231)	0.460*** (0.0388)	0.186*** (0.0475)	-0.135*** (0.0500)	0.346*** (0.124)	0.311*** (0.0389)				
Market Leverage	0.187 (0.159)	0.682** (0.295)	0.384 (0.456)	1.401*** (0.364)	1.405 (1.031)	1.572*** (0.357)	0.204 (0.199)	0.689* (0.294)	0.489 (0.452)	1.411*** (0.364)	1.357 (1.027)	1.622*** (0.354)				
Market-to-Book	0.00258 (0.0242)	0.113*** (0.0116)	0.0376 (0.0379)	-0.0666 (0.0720)	-1.117*** (0.388)	-0.365*** (0.177)	0.00368 (0.0242)	0.114*** (0.0125)	0.0480 (0.0571)	-0.0655 (0.0720)	-1.101*** (0.381)	-0.365*** (0.119)				
Fixed-Assets Ratio	-0.239* (0.144)	0.607*** (0.182)	-0.467 (0.403)	1.191*** (0.324)	1.165** (0.549)	0.600** (0.282)	-0.245* (0.144)	0.606*** (0.182)	-0.478 (0.401)	1.187*** (0.324)	1.207** (0.545)	0.581** (0.283)				
Cash Flow	-0.184 (0.382)	0.535 (0.735)	-2.072*** (0.594)	-2.115*** (0.614)	2.777 (2.447)	4.378*** (1.256)	-0.161 (0.384)	0.582 (0.735)	-1.855*** (0.587)	-2.180*** (0.613)	2.596 (2.421)	4.516*** (1.251)				
Cash	-1.953*** (0.325)	-4.935*** (0.661)	1.264** (0.509)	0.0375 (0.910)	3.031** (1.372)	-2.082*** (0.801)	-1.954*** (0.325)	-4.963*** (0.660)	2.097* (0.515)	0.0270 (0.911)	3.039** (1.344)	-2.151*** (0.807)				
Debt Rating Dummy	-0.0710 (0.0501)	-0.0341 (0.0909)	-0.372** (0.183)	0.181 (0.148)	0.268* (0.152)	0.158 (0.129)	-0.0708 (0.0500)	-0.0339 (0.0909)	-0.375** (0.178)	0.180 (0.140)	0.270* (0.154)	0.157 (0.126)				
Sales Growth	0.483*** (0.0714)	0.467*** (0.0669)	0.616*** (0.178)	0.851*** (0.139)	1.078*** (0.280)	0.678*** (0.199)	0.478*** (0.0710)	0.462*** (0.0669)	0.617*** (0.172)	0.849*** (0.140)	1.106*** (0.287)	0.546*** (0.119)				
Stock Returns	0.0388 (0.0388)	0.0611 (0.0611)	0.0639 (0.0639)	0.0818 (0.0818)	0.173 (0.173)	0.0948 (0.0948)	0.0395 (0.0395)	0.0607 (0.0606)	0.0621 (0.0621)	0.0817 (0.0817)	0.170 (0.170)	0.0965 (0.0965)				
Term Spread	0.0286** (0.0131)	0.0031*** (0.0020)	0.231*** (0.0551)	0.334*** (0.0529)	-0.168 (0.155)	-0.0120 (0.0421)	0.0282* (0.0135)	0.100*** (0.0212)	0.267*** (0.0516)	0.339*** (0.0532)	-0.170 (0.165)	-0.00699 (0.0414)				
Recession Dummy	0.0635** (0.0457)	0.134** (0.0648)	1.039*** (0.149)	0.185 (0.200)	-1.320 (1.023)	0.405*** (0.124)										
Low Growth Dummy							-0.0505 (0.0364)	0.0758* (0.0430)	0.453*** (0.113)	0.105 (0.134)	0.157 (0.352)	-0.0160 (0.0850)				
Constant	-4.582*** (0.193)	-7.542*** (0.321)	-7.155*** (0.420)	-5.425*** (0.467)	-9.428*** (1.131)	-7.509*** (0.365)	-4.562*** (0.193)	-7.559*** (0.322)	-7.180*** (0.421)	-5.438*** (0.485)	-9.573*** (1.155)	-7.644*** (0.367)				
Observations	112,512								112,512							
Pseudo R ²	0.044								0.044							

This table reports coefficient estimates for a multinomial logit model. The dependent variable includes six different types of security issuance: bank loan, public bond, convertible debt, SEO, and private placements of equity and debt. The base outcome is not issuing any type of security. Variables are defined in Appendix Table 1. The sample period is from 1988 to 2007. Panel A is for all firms; Panel B is for noninvestment-grade firms only; and Panel C is for investment-grade firms only. We define noninvestment-grade firms as those that never issued an investment-grade public bond during the entire sample period based on Moody's ratings. All regressions, except for the ones in Panel C, include industry fixed effects. Standard errors, corrected for clustering of observations at the firm level, are in parentheses. The symbols ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

5.6 Figure 3 and Table 6: credit spreads and cash use of proceeds

Read:

- Figure 3 shows that credit spreads widen sharply in recessions, especially around 1974, 1980–1982, 1990, 2001, and 2007.
- Wider spreads imply that risky securities become more expensive relative to safe securities.
- Table 6 estimates whether capital raised in recessions is saved as cash and short-term investments.
- Investment-grade firms save more proceeds as cash during recessions, especially from all financing sources and public debt.
- Noninvestment-grade firms are less able to accumulate cash from financing in downturns, consistent with tighter access and more immediate funding needs.

Economic message: The use-of-proceeds evidence supports a precautionary-liquidity interpretation for high-quality firms: they raise funds when available and save them during downturns.

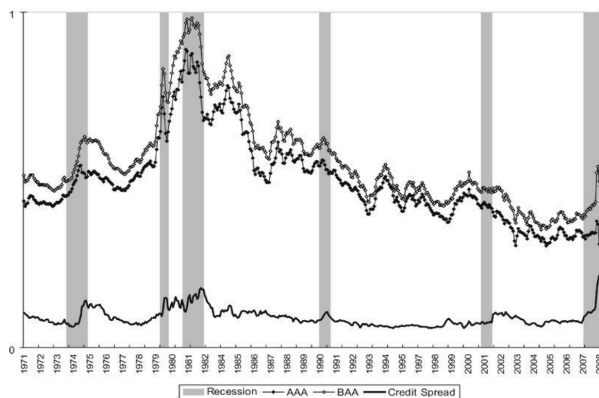


Figure 3
Credit spread over time
This figure presents the yields of AAA and BAA corporate bonds as well as the spread between the two for each calendar month from 1971 to 2008. The shaded areas correspond to recessions as defined by the NBER.

Figure 3: credit spreads over time

5.7 Table 7: maturity and security of public bonds and bank loans

Read:

- Panel A studies public bond issues and asks whether bonds are short-term and whether they are secured.
- Recession, low-growth, and weak-credit indicators increase the probability that public bonds are short-term.
- The results for secured public bonds are more mixed, although low-growth periods raise the probability of secured bonds in some specifications.
- Panel B studies bank loans and finds that downturn and credit-tightness measures increase the probability of short-term loans.
- Weak credit supply and low GDP growth also increase the probability that bank loans are secured.

Economic message: Debt contracts become safer during adverse macroeconomic conditions: maturities shorten, and bank loans are more likely to be secured.

Table 6
Cash holdings from capital-raising activities

Dependent Variable: Cash and		Speculative Grade Firms				Investment Grade Firms			
		β_1		β_2		β_1		β_2	
Short-term Investments		Coeff	t	Coeff	t	Coeff	t	Coeff	t
All Sources	1Q	0.318	101.48	-0.077	-7.14	0.089	18.42	-0.015	-1.17
	4Q	0.324	69.01	-0.064	-4.70	0.069	12.43	0.050	3.30
	8Q	0.285	50.45	-0.086	-5.36	0.084	13.61	0.079	4.66
SEOs	1Q	0.917	103.31	-0.013	-0.57	0.822	28.48	0.008	0.12
	4Q	0.816	66.95	0.027	0.84	0.653	14.14	0.869	8.62
	8Q	0.647	38.32	-0.005	-0.12	0.716	14.34	0.471	4.35
CBs	1Q	0.637	27.15	0.152	2.83	0.275	8.44	-0.101	-2.16
	4Q	0.568	19.35	0.029	0.42	0.187	4.85	-0.055	-1.00
	8Q	0.525	13.93	-0.139	-1.60	0.168	3.20	-0.061	-0.82
Bonds	1Q	0.475	38.87	-0.145	-3.81	0.076	11.30	-0.029	-2.22
	4Q	0.472	36.13	-0.160	-3.84	0.102	10.52	-0.004	-0.20
	8Q	0.485	29.66	-0.166	-3.20	0.116	9.36	0.093	3.81
Loans	1Q	0.097	28.64	-0.057	-5.41	0.087	10.41	-0.019	-0.65
	4Q	0.072	13.94	-0.054	-3.40	0.053	6.25	0.022	0.80
	8Q	0.074	12.06	-0.075	-4.06	0.062	6.74	0.030	1.01
Private Placements: Equities	1Q	0.688	52.52	-0.187	-6.35	0.378	4.18	-1.239	-3.98
	4Q	0.621	30.41	-0.146	-3.16	1.303	7.98	-0.223	-0.40
	8Q	0.530	20.66	-0.139	-2.46	0.878	6.98	-0.624	-1.46
Private Placements: Debt	1Q	0.161	18.18	-0.064	-2.42	0.078	4.84	0.004	0.11
	4Q	0.097	8.86	-0.070	-2.06	0.031	1.36	-0.134	-2.61
	8Q	0.094	6.23	-0.125	-2.80	0.014	0.50	0.288	4.55

This table presents the estimation results from the following regression specification:

$$Y = \beta_0 + \beta_1 \ln \left[\left(\frac{\text{capital raised}}{\text{total assets}_0} \right) + 1 \right] + \beta_2 \ln \left[\left(\frac{\text{capital raised}}{\text{total assets}_0} \right) + 1 \right] \\ \times \text{RecessionDummy} + \beta_3 \ln [\text{total assets}_0] + \varepsilon,$$

where $Y = \ln[(V_t - V_0)/\text{total assets}_0 + 1]$, and $V =$ quarterly cash and short-term investments. $t = 1, 4, 8$ corresponds to the fiscal quarter following the issuing quarter. In the first three rows, all new issues regardless of type of security are aggregated within a calendar quarter and these quarters are matched with the NBER's expansion/recession dates. In the remaining rows, we estimate the result separately for each of the security types. We report the results separately for noninvestment- and investment-grade firms. Noninvestment-grade firms are defined as those that never issued an investment-grade public bond during the entire sample period.

down cycles, investment-grade firms save substantially more from all financing sources as a whole and especially from SEOs and public bonds over the

Table 6: cash holdings from capital raising

Table 7
Factors affecting the maturity and security of public bonds and bank loans: Logit model

Panel A: Public Bond Issues					
	Short-term vs. Long-term Bond			Secured vs. Unsecured Bond	
	(1)	(2)	(3)	(4)	(5)
Firm Age	0.0166*** (0.00489)	0.0159*** (0.00487)	0.0183*** (0.00489)	-0.00681 (0.0124)	-0.00666 (0.0130)
In(Total Assets)	0.275*** (0.0396)	0.275*** (0.0396)	0.254*** (0.0420)	-0.244** (0.0972)	-0.543*** (0.120)
Leverage	0.183 (0.516)	0.205 (0.509)	0.199 (0.575)	4.471*** (0.679)	4.482*** (0.681)
Market-to-Book	0.218*** (0.0571)	0.228*** (0.0565)	0.186*** (0.0621)	-0.185 (0.270)	-0.180 (0.263)
Fixed-Assets Ratio	-0.765*** (0.352)	-0.772** (0.351)	-0.766** (0.382)	2.987*** (0.790)	3.024*** (0.789)
Cash Flow	-1.431** (0.668)	-1.414** (0.658)	-1.302* (0.680)	-0.591 (1.131)	-0.652 (1.099)
Cash	-1.615* (0.858)	-1.714** (0.858)	-1.524* (0.871)	2.688** (1.066)	2.879*** (1.061)
Inverse Interest Coverage	-0.00550 (0.149)	-0.00796 (0.147)	0.00675 (0.148)	-0.0682 (0.154)	-0.0761 (0.156)
Debt Rating Dummy	-0.343 (0.232)	-0.335 (0.232)	-0.510* (0.293)	-0.270 (0.203)	-0.178 (0.212)
Sales Growth	-0.376* (0.195)	-0.390** (0.196)	-0.490** (0.223)	0.160 (0.197)	0.159 (0.189)
Stock Return	-0.384*** (0.139)	-0.384*** (0.139)	-0.249** (0.125)	-0.566*** (0.138)	-0.527*** (0.134)
Term Spread	-15.38*** (4.667)	-13.55*** (4.622)	-13.85*** (5.038)	6.016 (8.991)	8.957 (8.753)
Recession Dummy	0.405*** (0.124)			-0.362 (0.228)	
Low Growth Dummy		0.148* (0.0844)			0.373** (0.157)
Weak Credit Dummy			0.430*** (0.102)		0.123 (0.203)
Constant	-3.676*** (0.553)	-3.696*** (0.550)	-3.632*** (0.603)	-4.489*** (1.181)	-4.597*** (1.193)
Observations	7,523	7,523	6,664	8,314	8,314
Pseudo R ²	0.111	0.110	0.120	0.250	0.251

(continued)

Table 7
Continued
Panel B: Bank Loans

	Short-term vs. Long-term Bond			Secured vs. Unsecured Bond	
	(1)	(2)	(3)	(4)	(5)
Firm Age	0.00224 (0.00214)	0.00211 (0.00214)	0.00237 (0.00220)	-0.0143*** (0.00319)	-0.0146*** (0.00320)
In(Total Assets)	-0.0569*** (0.0205)	-0.0566*** (0.0205)	-0.0497** (0.0211)	-0.778*** (0.0327)	-0.784*** (0.0330)
Leverage	0.0427 (0.134)	0.729 (0.133)	-0.111 (0.138)	3.498*** (0.232)	3.497*** (0.233)
Market-to-Book	0.0693*** (0.0237)	0.0722*** (0.0238)	0.0517** (0.0238)	-0.0506** (0.0257)	-0.0528** (0.0256)
Fixed-Assets Ratio	-0.227 (0.150)	-0.230 (0.150)	-0.181 (0.155)	-0.688*** (0.228)	-0.686*** (0.228)
Cash Flow	-1.564*** (0.287)	-1.555*** (0.285)	-1.687*** (0.279)	-3.366*** (0.512)	-3.360*** (0.516)
Cash	-0.0667 (0.210)	-0.0857 (0.210)	0.0257 (0.218)	0.721** (0.327)	0.721** (0.327)
Inverse Interest Coverage	-0.0947** (0.0426)	-0.0941** (0.0426)	-0.0804* (0.0450)	0.137** (0.0682)	0.139** (0.0684)
Debt Rating Dummy	-0.403*** (0.0663)	-0.406*** (0.0663)	-0.462*** (0.0687)	0.451*** (0.0960)	0.463*** (0.0960)
Sales Growth	-0.195*** (0.0476)	-0.209*** (0.0476)	-0.200*** (0.0495)	0.129 (0.0807)	0.141* (0.0816)
Stock Return	-0.105*** (0.0242)	-0.108*** (0.0243)	-0.0872*** (0.0245)	0.0923** (0.0395)	0.0787* (0.0398)
Term Spread	26.80*** (1.811)	28.78*** (1.860)	29.18*** (1.928)	1.069 (2.675)	3.220 (2.725)
Recession Dummy	0.655*** (0.0839)			0.0961 (0.109)	
Low Growth Dummy		0.102** (0.0437)			0.302*** (0.0701)
Weak Credit Dummy			0.599*** (0.0397)		0.130** (0.0569)
Constant	0.823** (0.336)	0.815** (0.344)	0.677* (0.366)	4.856*** (0.631)	4.807*** (0.629)
Observations	15,356	15,356	14,627	10,149	10,149
Pseudo R ²	0.0634	0.0602	0.0741	0.296	0.296

This table reports coefficient estimates for a logit model. Panel A includes only public bond issuances and their characteristics from 1985 to 2007 in the first three columns (since there is no short-term bond issue before 1985 in our sample) and from 1971 to 2007 in the last three columns. Panel B includes bank loans only, and the sample period is from 1988 to 2007. In columns 3 and 6 of both Panels A and B, where we include weak credit dummy, the sample period is from the second quarter of 1990–2007. The dependent variable is equal to one if the public debt or bank loan is short term in columns 1–3, or secured in columns 4–6. All regressions include industry fixed effects. Standard errors, corrected for clustering of observations at the firm level, are in parentheses. The

Table 7, Panel A: public bonds

Table 7, Panel B: bank loans

5.8 Appendix Table A1 and Table A2: variable definitions and robustness

Read:

- Appendix Table A1 defines all variables, including capital-raising proceeds, debt maturity, secured status, firm controls, and macroeconomic dummies.
- Appendix Table A2 repeats the multinomial logit using the weak-credit dummy as the downturn measure.
- The weak-credit robustness confirms that financing choices respond not only to recessions or GDP growth, but also to direct survey evidence of loan-supply tightening.

Economic message: The appendix reinforces the interpretation that credit-market conditions, not only real-side business-cycle conditions, shape capital raising.

Appendix

Table A1

Variable	Definition	Source
Bond Issuance Proceeds	Total proceeds raised through a public bond offering in a given month	Mergent FISD
Bond Maturity Dummy	Set equal to one if the proceeds-weighted initial maturity of bonds issued in a given month is less than five years and zero otherwise	Mergent FISD
Bond Security Dummy	Set equal to one if the proceeds-weighted secured bond dummy in a given month is greater than 0.5 and zero otherwise	Mergent FISD
Cash	Cash and short-term investments, scaled by total assets	Compustat
Cash Flow	Income before extraordinary items plus depreciation, scaled by the book value of total assets	Compustat
Convertible Bond Proceeds	Total proceeds raised through a convertible bond offering in a given month	Mergent FISD
Credit Quality	Obtained from Moody's credit ratings and classified as follows: 0 - not rated, 1 - C to Aaa1, 2 - B3 to Baa1, 3 - Baa3 to Baa1, 4 - A3 to Aaa	Mergent FISD
Equity Issuance Proceeds	Total proceeds raised through a seasoned equity offering in a given month	SDC Global
Financing Choice Variable for the Multinomial Logit Models	Classified as following for each firm-month: 0 - No issue, 1 - Loan, 2 - Bond, 3 - Convertible, 4 - Seasoned equity offering, 5 - Private placement of equity, 6 - Private placement of debt. In months with multiple issues, the classification is determined by the largest issue in terms of proceeds raised	Dealscan, Mergent FISD, SDC Global

(continued)

Table A1 Continued

Variable	Definition	Source
Fixed Asset Ratio	Net property, plant and equipment scaled by the book value of total assets	Compustat
Inverse Interest Coverage	$\log(1 + (\text{Interest Expense}/\text{EBIT}))$	Compustat
Loan Maturity Dummy	Set equal to one if the proceeds-weighted initial maturity of loans obtained in a given month is less than five years and zero otherwise	Dealscan
Loan Proceeds	Total proceeds raised through a bank loan in a given month	Dealscan
Loan Security Dummy	Set equal to one if the proceeds-weighted secured loan dummy in a given month is greater than 0.5 and zero otherwise	Dealscan
Log(Total Assets)	Natural logarithm of the book value of assets in constant 1994 dollars	Compustat
Low Growth Dummy	Set equal to one in quarters in which GDP growth was below the 25th percentile of growth between 1971 and 2007 and zero otherwise	BEA
Market leverage	Long-term debt plus debt in current liabilities, scaled by the market value of assets (total assets - book value of equity + market value of equity)	Compustat
Market to Book	Book value of total debt plus the liquidating value of preferred stock plus the market value of equity, scaled by the book value of total assets	Compustat
Rated Firm Dummy	Indicator set equal to 1 if a firm has an S&P domestic long-term issuer credit rating and zero otherwise	Compustat
Recession Dummy	Set equal to one in months designated as recession by the NBER	NBER
Sales Growth	Percentage change in sales over the previous year	Compustat
Secured Bond Dummy	Set equal for to one if an issued bond is classified as secured	Mergent FISD
Secured Loan Dummy	Set equal for to one if a bank loan is classified as secured	Dealscan
Stock Return	Previous twelve-month stock return	CRSP
Term Spread	Difference in the yields on ten-year treasuries and one-year treasuries	Federal Reserve
Weak Credit Dummy	Set equal to one in months when the net percentage of senior loan officers tightening standards for large to medium firms is positive and zero otherwise	Federal Reserve

This data appendix describes the primary variables of interest. All firm characteristics, unless noted otherwise, represent beginning-of-year values. Data sources, provided in the last column, include Compustat, CSRP, Loan Pricing Corporation's DealScan, Mergent Fixed Income Securities Database, SDC Global New Issues Database, U.S. Bureau of Economic Analysis, National Bureau of Economic Research, and the U.S. Federal Reserve Board.

Appendix Table A1, part 1: variable definitions

Appendix Table A1, part 2: variable definitions

Table A2
A multinomial logit model of security choice using weak credit dummy for downturns

	Panel A						Panel B						Panel C					
	Loan	Bond	Convert	SE0	Private Placements	Equity	Loan	Bond	Convert	SE0	Private Placements	Equity	Loan	Bond	Convert	SE0	Private Placements	Equity
Firm Age	-0.0040***	-0.0036***	-0.0188***	-0.0444***	-0.0230***	-0.0060***	-0.0062***	-0.0113***	-0.0071***	-0.0248***	-0.0062***	0.0008***	0.0032	-0.0008	-0.0264***	-0.0264***	-0.0008***	-0.0008***
Ln(Total Assets)	0.0000	0.0022	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Market Leverage	-0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Market to Book	-0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Fixed Assets Ratio	-0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cash Flow	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Cash	-0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Debt Rating Coverage	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Debt Rating Dummy	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Sales Growth	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Stock Return	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Term Spread	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Weak Credit Dummy	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000
Constant	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	65,337						100,107						65,337					
Pseudo R ²	0.00						0.00						0.00					

This table reports coefficient estimates for a multinomial logit model as in Table 5 but using weak credit dummy for downturns. The dependent variable includes six different types of security issuance: bank loan, public bond, convertible debt, SE0, and private placements of equity and debt. The base outcome is not issuing any type of security. Variables are defined in Appendix Table 1. The sample period is from the second quarter of 1990-2007. Panel A is for all firms; Panel B is for noninvestment-grade firms only; and Panel C is for investment-grade firms only. A Wald test is conducted for the null hypothesis that the coefficients on the variables in the first four rows are equal to zero.

6 Short summary

- The paper studies corporate capital raising across public debt, equity, convertibles, bank loans, and private placements.
- The sample covers U.S. industrial firms from 1971 to 2007, with later start dates for loans and private placements.
- Aggregate capital raising is not uniformly procyclical: SEOs are procyclical, but public debt can rise in downturns.
- The composition of public debt shifts toward higher-rated bonds during recessions and low-growth periods.
- Noninvestment-grade firms are much more constrained during downturns; their equity and low-quality debt issuance decline.
- Investment-grade firms can raise public debt countercyclically.

- Firm characteristics matter: public debt issuers are larger and safer, while equity issuers tend to be younger and have higher market-to-book ratios.
- Investment-grade firms often raise capital during downturns and save proceeds as cash.
- Debt contracts become safer in adverse conditions, especially through shorter maturity and secured bank loans.
- The evidence is consistent with supply-side channels such as credit crunches and flight to quality, along with some demand-side effects.

7 Conclusion

7.1 Core conclusion

Erel, Julio, Kim, and Weisbach show that macroeconomic conditions strongly affect firms' access to external finance and the securities they use. Poor conditions do not simply lower total financing for all firms. Instead, they reallocate capital toward safer borrowers and safer securities.

7.2 Economic intuition

When macroeconomic conditions deteriorate, investors and intermediaries become less willing to fund risky borrowers. Low-quality firms face reduced access to public equity, low-rated debt, and loans. High-quality firms, by contrast, can still access public bond markets and may raise funds opportunistically to build liquidity. Investors' demand shifts toward safe claims, and debt contracts become shorter and more secured.

7.3 Takeaway

The paper provides broad evidence that capital structure and security design are shaped by macro-financial conditions. Credit quality determines whether downturns shut firms out of markets or allow them to raise precautionary financing. The central lesson is that financial flexibility is valuable because access to capital is state-contingent and uneven across firms.